

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th July 2022

Subject SP5–Investment and Finance

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** Consider a 10-year Interest Rate Swap (IRS) based on a notional principal of 100 million. Under the terms of the swap, Company A agrees to make interest payments annually in arrears based on a fixed interest rate in return for which Company B (or C or D) makes corresponding floating interest rate payments based on the 1-year spot Inter Bank Offer Rate (IBOR). The investment bank arranging the swap needs 5 bps (0.05%) p.a. of notional principal as its fee.

Below table shows interest rates in the market at which Company A, B, C and D can issue bonds based on their credit rating profile:

Company	Fixed rate	Floating rate
A	9% p.a.	IBOR + 0.60%
B	8% p.a.	IBOR – 0.25%
C	10% p.a.	IBOR + 1.55%
D	11% p.a.	IBOR + 2.45%

- i) Should Company A swap interest cash flows with Company B, C or D? Justify your answer. (4)
- ii) Define credit risk and mention two ways to control it. (2)
- iii) Discuss how the credit risk exposure of an IRS will be expected to change over the life of the swap for the receiver of the fixed rate when the yield curve is downward sloping. Assume that interest rates will evolve as implied by current yield curve. (3)
- iv) In the context of mean-variance portfolio theory, explain briefly the following terms:
- opportunity set
 - efficient frontier
 - indifference curves
 - optimum portfolio

(4)
[13]

- Q. 2)** ABC Ltd. is a non-dividend paying company whose current share price is 1000. The following table shows the current market prices of four exchange traded European-style call options available on ABC shares.

S. No.	Strike Price	Term to Maturity	Premium
1	925	1 month	85
2	1000	1 month	30
3	1050	1 month	10
4	1000	2 months	45

An options trader has one-month investment horizon. At the month end, he will close out his positions in any call options (long or short) available in his portfolio.

Strategy A: The trader intends to buy one call with a strike price of 1050 and sell one call with a strike price of 1000 both having one month to expiry.

Strategy B: The trader intends to buy one call with a strike price of 1000 having two months to expiry and sell one call with a strike price of 1000 having one month to expiry.

Strategy C: The trader takes the following positions

- buys one call with a strike price of 1050 having one month to expiry
- sells two calls with a strike price of 1000 having one month to expiry
- buys one call with a strike price of 1000 having two months to expiry

For the first two parts of this question, your answer should include statements / comments on

- the level (or range) of ABC share price where the strategy is profitable or loss making or breaks-even; consider all break-even points
- the strategy's maximum profit and maximum loss
- the trader's views on ABC share price movements in coming month

i) Explain briefly, when strategy A might make sense. (3)

ii) Discuss briefly, when strategy B might make sense. (6)

iii) Identify strategy C and state the trader's view on ABC share price. (2)

Finally, the trader chooses strategy C. Within two days of implementing Strategy C, new information came into the market about ABC shares, which is expected to have significant impact (i.e. more than 5%) on ABC share price.

iv) State the adjustments required to Strategy C on the day of the news, buying / selling only the last two tabulated call options (i.e. call options with premium 10 and 45), if the new information is bad news and ABC share price fell down to 980. (2)
[13]

Q. 3) In a developing country ABC, twenty years ago, insurance sector was opened up for private players that have become profitable from past few years. Currently many life insurers are coming up for Initial Public Offerings (IPOs) and their major task is to set an appropriate value of the company.

The above country (ABC) was dependent upon two other countries for importing crude oil and coal. A war broke out between those two countries – hence severely impacting country ABC's crude oil and coal imports. This shook the economy of ABC and inflation increased significantly. The central bank of ABC immediately raised base interest rate by 0.50% p.a. in order to control inflation.

The Government intends to reduce dependency on crude oil and other fossil fuels that increase the economy's carbon emissions in to the atmosphere. Carbon emissions include gases like carbon dioxide (CO₂), methane (CH₄) etc. that causes global warming.

Along with a host of countries, the Government has recently announced major commitments to significantly cut carbon emissions with an aim to reach "net zero" carbon footprint in the coming years.

i) Discuss the challenges in valuing life insurance companies as compared to valuing companies from other industries using fundamental analysis and tools like Price to earnings ratio. (4)

ii) Suggest various ways to value life insurance companies. Your answer should include key metrics for comparison that will help a prospective investor in choosing one life insurer over another life insurer. (5)

- iii)** Mention other ways to control inflation by the Government / central bank in addition to the 0.50% p.a. increase in interest rate. (4)
- iv)** Suggest possible ways to reduce the net carbon emissions to zero by considering environmental issues (ESG) at an individual level as well as at the Government level with more emphasis on individual level. (6)
- v)** In the context of behavioural finance, write the name of the behaviour / property displayed in following cases:
- a) Top five equity mutual funds have more than 50% of their funds invested in these 15 stocks. I will also invest in these 15 stocks.
 - b) Today is full moon day, so market should move up as this happened many times in the past. I will buy call option on the equity index.
 - c) Today my favourite equity analyst in business TV channel said that Banking sector would perform well. I will buy Bank stocks.
 - d) During the past month, these three stocks came more than ten times in the list of daily top five gainers. I will buy these three stocks.
 - e) Gold prices are rising every month for the past six months. I will buy gold ETF (exchange traded funds).

(5)
[24]

- Q. 4) i)** Explain how a company's corporate dividend policy might be affected by the tax system. (3)
- ii)** Outline the factors an investor should consider when deciding an investment strategy to maximise net of tax returns. (4)

[7]

- Q. 5)** In the nation of Actuarial, inflation and interest rates used to be far higher in the 1990s compared with most developed nations. Many insurance companies had issued contracts in the past 20-30 years which included guaranteed annuity rates irrespective of interest rate movements. However, since the year 2000, interest rates have been falling. In fact, the interest rates are so low that most of these guarantees have started to bite, causing financial challenges to the insurance companies.

Describe how the ideas of behavioural finance can be used to explain this situation. [5]

- Q. 6) i)** Explain in detail what asset pricing models are used for and the main types of asset pricing models. (6)
- ii)** The trustees of a pension scheme would like to hedge the longevity risk associated with the scheme's pensions in payment. The suggestion is to invest in a longevity bond that offers coupons linked to survival according to a life table for annuitants. The bond would be issued by an investment bank solely to the pension scheme.

Outline the advantages and disadvantages of hedging the longevity risk. (6)

iii) A Global equity unit trust manages its portfolio as follows:

- 75% of the fund is invested passively to track a global equities benchmark
- 25% of the fund is actively invested in a range of equities

Give possible reasons why the fund may have been structured in the above way.

(6)

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Q. 7) i) You have been asked to design a suitable investment strategy for a charitable fund that awards scholarships each year to promising young Actuaries and Accountants.

Describe the process of risk budgeting and explain how it could help you to achieve the above task.

(8)

ii) You are also advising the fund to consider the risk-adjustment performance of two trusts, one of which can be the model investment strategy for the fund. You have been given the following information on the performance of the two investments and the benchmark index over the last five years.

The risk-free rate of return over the period was 1%.

Calculate **four** different risk-adjusted performance measures for each fund.

(5)

	Trust A	Trust B	Benchmark Index
Average Return	4% pa	5% pa	2% pa
Standard Deviation	55% pa	25% pa	15% pa
Correlation Coefficient with Benchmark	0.33	0.87	1.00

iii) Discuss which of the two investment trusts the fund should choose to invest in, on the basis of your calculations in (ii), noting any assumptions and limitations.

(7)

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